Introduction to Microeconomics Fall 2025

SEMINAR #2

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Demand elasticities

- Three measures of demand elasticity:
  - Own-price elasticity of demand → the percentage change in the quantity demanded resulting from a percent increase in the price of the good
  - Cross-price elasticity of demand → percentage change in demand of good *x* due to a percentage change in the price of good *y*
  - Income elasticity of demand → percentage change in demand of good due to a percentage change in income

Own-price elasticity of demand: formula

- By definition, the own-price elasticity of demand, ε_P^D , is given by the ratio between the percentage change in the quantity demanded divided by the percentage change in price:

$$\text{Own– price elasticity of demand} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

- Hence, in algebraic terms:

$$\varepsilon_P^D = \frac{\%\Delta Q^D}{\%\Delta P} = \frac{\Delta Q^D / Q^D}{\Delta P / P}$$

where the % variation is calculated with respect to the initial value of Q^D and P, hence:

$$\varepsilon_P^D = \frac{\Delta Q^D / Q^D}{\Delta P / P} = \frac{(Q_1^D - Q_0^D)/Q_0^D}{(P_1 - P_0)/P_0}$$

Cross-price elasticity of demand

➢The cross-price elasticity of demand, $\varepsilon_{P_y}^{D_x}$, tells you by how much the quantity demanded of one good changes when the price of another good changes:

$$\text{Cross– price elasticity of demand} = \frac{\text{percentage change in quantity demanded of good x}}{\text{percentage change in price of good y}}$$

➢Hence, in algebraic terms:

$$\varepsilon_{P_y}^{D_x} = \frac{\%\Delta Q_x^D}{\%\Delta P_y} = \frac{\Delta Q_x^D / Q_x^D}{\Delta P_y / P_y}$$

➢If $\varepsilon_{P_y}^{D_x} > 0$, goods x and y are substitutes. If instead $\varepsilon_{P_y}^{D_x} < 0$, goods x and y are complements. [NOTE: **both the sign and magnitude are important here!** => we cannot take the absolute value]

Income elasticity of demand

➢The income elasticity of demand, ε_R^D, tells you by how much the quantity demanded of one good changes when income changes:

$$\text{Income elasticity of demand} = \frac{\text{percentage change in quantity demanded of good x}}{\text{percentage change in income}}$$

➢Hence, in algebraic terms:

$$\varepsilon_R^D = \frac{\%\Delta Q^D}{\%\Delta R} = \frac{\Delta Q^D / Q^D}{\Delta R / R}$$

➢If $\varepsilon_R^D > 0$, good x is a normal good. If instead $\varepsilon_R^D < 0$, good x is an inferior good. [NOTE: **both the sign and magnitude are important here!** => we cannot take the absolute value] ⟶ More on later.

Price elasticity of supply

- The price elasticity of supply, ε_P^S , is given by the ratio between the percentage change in the quantity supplied divided by the percentage change in price:

$$\text{Price elasticity of supply} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

- Hence, in algebraic terms:

$$\varepsilon_P^S = \frac{\%\Delta Q^S}{\%\Delta P} = \frac{\Delta Q^S / Q^S}{\Delta P / P}$$

where the % variation is calculated with respect to the initial value of Q^S and P, hence:

$$\varepsilon_P^S = \frac{\Delta Q^S / Q^S}{\Delta P / P} = \frac{\left(Q_1^S - Q_0^S\right) / Q_0^S}{\left(P_1 - P_0\right) / P_0}$$

- What is the sign of the price elasticity of supply?

Exercise 1

The cross elasticity of demand between cinema tickets and online streaming services is defined as

a) the price elasticity of demand for cinema tickets divided by the price elasticity of demand for online streaming services.

b) the change in the quantity of cinema tickets demanded divided by the change in the quantity of online streaming services demanded.

c) the percentage change in the quantity of cinema tickets demanded divided by the percentage change in the price of online streaming services.

d) the percentage change in the quantity of cinema tickets demanded divided by the percentage change in the quantity of online streaming services demanded.

Exercise 2

From Geneva, SWISS offers direct flights to Barcelona for CHF 500.- (round trip). During the last 3 months, an average of 10 passengers have purchased tickets for each day. For the Christmas holidays, SWISS has just launched a round trip offer for CHF 300.- thanks to which flights are full (20 passengers per flight) during the holidays. What is the price elasticity of demand (in absolute value)?

a) 0.5

b) 2

c) 1

d) 2.5

$P_1 = 500$ $\quad Q^D_1 = 10$

$P_2 = 300$ $\quad Q^D_2 = 20$

$$\varepsilon^D_P = \frac{(20-10)/10 \cdot \cancel{100\%}}{(300-500)/500 \cdot \cancel{100\%}} = \frac{10/10}{-200/500} = -2.5$$

$$|\varepsilon^D_P| = 2.5$$

Exercise 3

Market research has shown that the demand function for soccer balls is as follows: $Q_D = 400 - 5P + 20P_B$, where P is the price of a soccer ball and P_B is the price of a basketball. The current price of a soccer ball is 20 and the price of a basketball is 15. At these price levels, what is the cross-price elasticity of demand for soccer balls with respect to the price of basketballs?

a) 0.25

b) 0.66

c) 0.5

d) 0.33

$$Q_D = 400 - 5P + 20P_B$$

$$\left.\begin{array}{l} P = 20 \\ P_B = 15 \end{array}\right\} \quad Q_D = 400 - 5 \cdot 20 + 20 \cdot 15 = 600$$

$$\left.\begin{array}{l} P = 20 \\ P_B = 16 \end{array}\right\} \quad Q_D = 400 - 5 \cdot 20 + 20 \cdot 16 = 620$$

$$\varepsilon^{D}_{P_B} = \frac{[(620-600)/600]\,\cancel{100\%}}{[(16-15)/15] \cdot \cancel{100\%}}$$

$$= \frac{20/600}{1/15} = 1/2$$

Exercise 4

Lilith spends all her income on comic books and ramen. Following the loss of her job, her income is cut in half and we see that her consumption of comics decreases and that of ramen increases. What can we deduce from this?

a) Comics and ramen are both normal goods for Lilith.

b) Comics are a normal good while ramen is an inferior good for Lilith.

c) Comics are a luxury good for Lilith and ramen is a necessity good for Lilith.

d) None of the above.

$$\varepsilon^D_R = \frac{(Q_1^D - Q_0^D)/Q_0^D \cdot 100\%}{(R_1 - R_0)/R_0 \cdot 100\%}$$

$R \downarrow \rightarrow \begin{cases} Q^D_{book} \downarrow & \varepsilon^D_R > 0 \\ Q^D_{ramen} \uparrow & \varepsilon^D_R < 0 \end{cases}$

$\rightarrow \begin{cases} \text{book: normal good} \\ \text{ramen: inferior good} \end{cases}$

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta R}$$

Income elasticity and types of goods (1)

- If $\varepsilon_R^D > 0$, good x is a normal good: an increase in income leads to an increase in demand.
- If instead $\varepsilon_R^D < 0$, good x is an inferior good: higher income leads to a decline in demand).
- Among normal goods we can also distinguish between:
  - Luxury goods ($\varepsilon_R^D > 1$) $\rightarrow$ the quantity demanded increases more than proportionally and the share of the good in the budget increases with increase in income.
  - Necessity goods ($0 < \varepsilon_R^D < 1$) $\rightarrow$ the quantity demanded increases less than proportionally to income, and the share of the good in the budget decreases with increase in income.

Exercise 5

Consider the demand for rental housing in San Francisco. The demand function is given by

Q_D = 1000 + 0.25R − P, with R = 4000 and P = 1600

1. Determine the quantity demanded.

2. Calculate the price elasticity of demand. Is demand elastic or inelastic?

3. Suppose rents increase by $100. Does total spending increase or decrease?

4. Suppose the price is $1000. Calculate the price elasticity of demand again. Is demand elastic or inelastic?

5. Suppose the price is $800. Calculate the price elasticity of demand again. Is demand elastic or inelastic? And how does the total expense vary when the rent increases by $100?

$$Q_D = 1000 + 0.25R - P$$

$$\begin{cases} R = 4000 \\ P = 1600 \end{cases}$$

1. $Q_D = 1000 + 0.25 . 4000 - 1600$

$= 400$

2. $P = 1601$

$Q_D = 1000 + 0.25 . 4000 - 1601$

$= 399$

$$\varepsilon_P^D = \frac{(399 - 400)/400}{(1601 - 1600)/1600}$$

$$= \frac{-1/400}{1/1600} = -4$$

$|\varepsilon_P^D| = 4 > 1 \Rightarrow$ demand is elastic

3. Price ↑ by 100 USD

Because demand is elastic, ↑ in price would lead to decrease in Quantity demanded more proportional

Total expenditure: $(P \times Q)$ ↓

P ↑, Q ↓

$P = 1600 + 100 = 1700$

$Q^D = 1000 + 0.25 \cdot 4000 - 1700$

$= 300$

Total expenditure before $= 400 \times 1600$

$= 640,000$

after $= 300 \times 1700$

$= 510,000$

4. $P = 1000$

$$Q^D = 1000 + 0.25 \cdot 4000 - 1000$$
$$= 1000$$

$P = 1001$

$$Q^D = 1000 + 0.25 \cdot 4000 - 1001$$
$$= 999$$

$$\varepsilon_P^D = \frac{(999 - 1000)/1000}{(1001 - 1000)/1000} = -1$$

$|\varepsilon_P^D| = 1 \rightarrow$ unit elasticity

$\rightarrow$ total expenditure is maximized

5. $P = 800 \rightarrow Q^D = 1000 + 0.25 \cdot 4000 - 800$
$$= 1200$$

$$\varepsilon_P^D = \frac{(1119 - 1200)/1200}{(801 - 800)/800} = -\frac{2}{3}$$

→ Demand is inelastic

Price ↑ by 100 USD, Quantity demanded decrease in smaller proportion

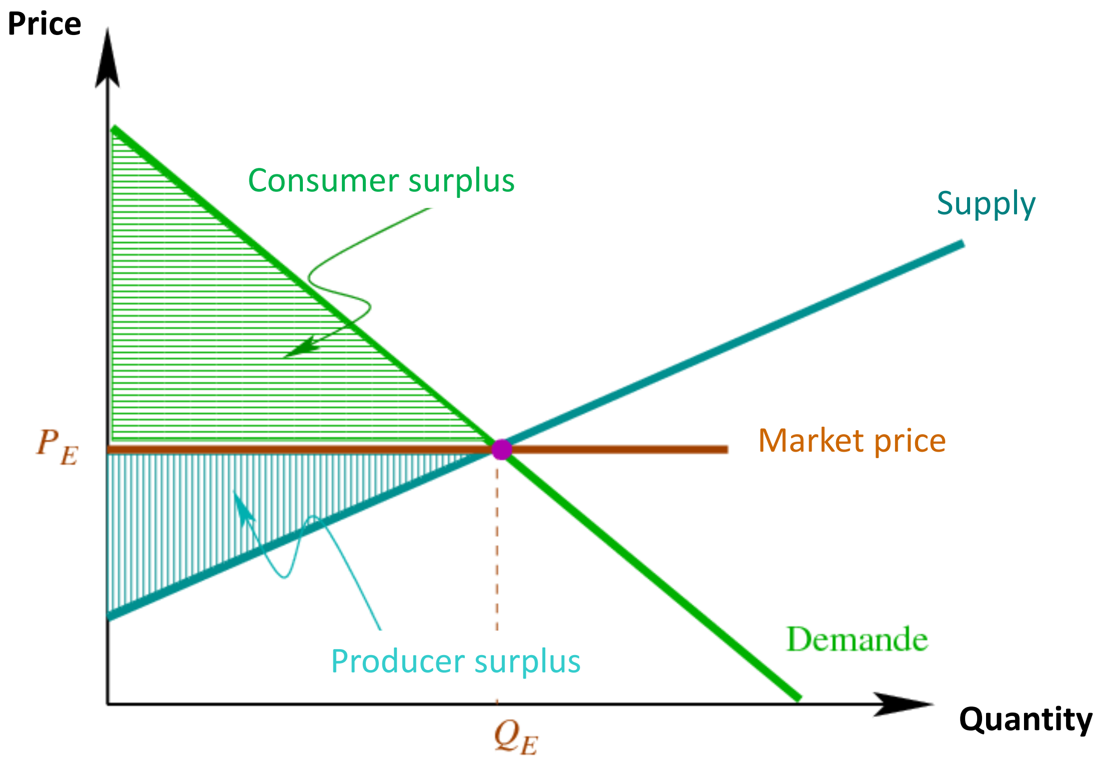
→ total expenditure: $(P \times Q)$ increases

$\uparrow \quad \downarrow$

$\uparrow$

Total surplus

➢ **Total surplus** will be given by the sum of consumer and producer surplus = buyers' willingness to pay - cost incurred by sellers



Consumer surplus	Buyers' willingness to pay — Price paid
+	+
Producer surplus	Price received by sellers — Cost incurred
=	=
Total surplus	Buyers' willingness to pay — Cost incurred

Exercise 6

$$\begin{rcases} Q_D = 120-2P \\ Q_S = 4P-48 \end{rcases} \begin{matrix} 120-2P = 4P-48 \\ 168 = 6P \rightarrow \begin{cases} P = 28 \\ Q = 64 \end{cases} \end{matrix}$$

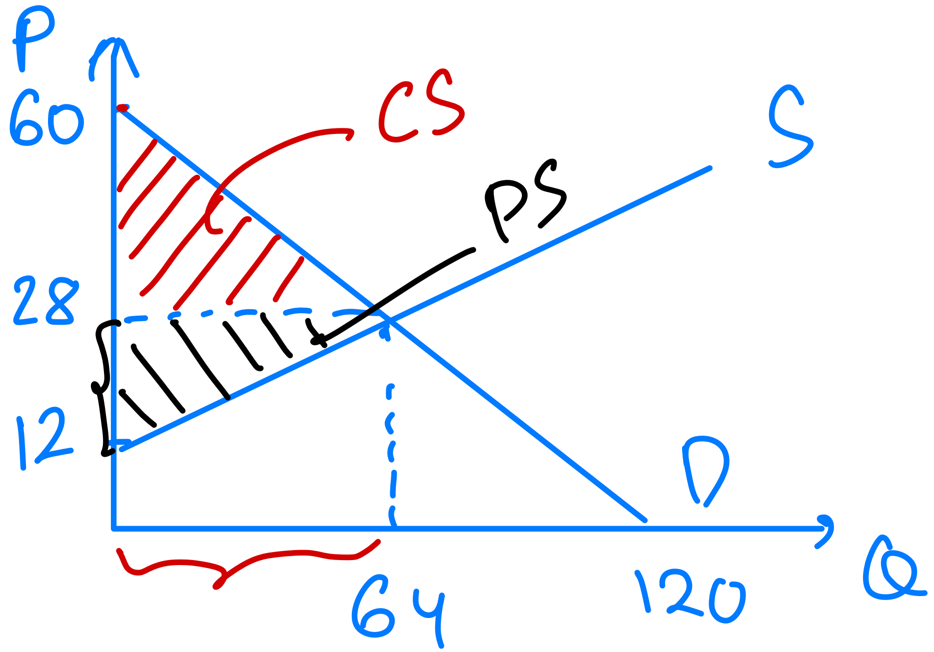
Consider the market for pencils characterized by the demand function Q_D = 120 − 2P and the supply function Q_S = 4P − 48 where Q is the quantity and P is the market price. What are the price P, quantity Q, consumer surplus CS and producer surplus PS at market equilibrium?

a) P = 28, Q = 64, SC = 2048, SP = 512.

b) P = 28, Q = 64, SC = 1024, SP = 512.

c) P = 64, Q = 28, SC = 1024, SP = 512.

d) P = 28, Q = 64, SC = 1025, SP = 1024.



$CS = (60-28) \cdot 64/2 = 1024$

$PS = (28-12) \cdot 64/2 = 512$

Exercise 7

Consider a market where, originally, the supply and demand functions are respectively: $QS1$ = 3P and QD = 20 - P. Following a technological change, the new supply function is: $QS2$ = 4P. Which of the following statements is correct?

a) Both before and after the technological change, the price elasticity of demand at equilibrium is equal to -1/3.

b) Both before and after the technological change, the price elasticity of supply at equilibrium is equal to 1.

c) After the technological change, the price elasticity of demand at equilibrium is equal to -4.

d) Before the technological change, the price elasticity of supply at equilibrium is equal to 3.

$$Q_S^1 = 3P \qquad Q_D = 20 - P$$

$$Q_S^2 = 4P$$

Before : $3P = 20 - P$

$$\Rightarrow 4P = 20 \rightarrow \boxed{\begin{array}{l} P^* = 5 \\ Q^* = 15 \end{array}}$$

$$P^* = 5 \rightarrow P = 6 \rightarrow Q_s^1 = 6 \cdot 3 = 18$$

$$Q_D = 20 - 6 = 14$$

$$\varepsilon_p^D = \frac{(14-15)/15}{(6-5)/5} = \frac{-1/15}{1/5} = -1/3$$

$$\varepsilon_p^S = \frac{(18-15)/15}{(6-5)/5} = \frac{3/15}{1/5} = 1$$

After:

$4P = 20 - P$

$\Rightarrow 5P = 20 \rightarrow \boxed{\begin{array}{l} P^* = 4 \\ Q^* = 16 \end{array}}$

$P^* = 4 \rightarrow P = 5$

$Q_S^2 = 5 \times 4 = 20$

$Q_D = 20 - 5 = 15$

$\varepsilon_P^D = \dfrac{(15-16)/16}{(5-4)/4} = \dfrac{-1/16}{1/4} = -1/4$

$\varepsilon_P^S = \dfrac{(20-16)/16}{(5-4)/4} = \dfrac{4/16}{1/4} = 1$

Exercise 8

Alsoft is a monopoly firm that provides highly specialized software. The unit price of its software is 75,000 francs. The willingness to pay (WTP) of potential customers is summarized by the table below:

Number of potential clients	WTP		
1	120,000	− 75,000	45,000
1	100,000	− 75,000	25,000
1	75,000	− 75,000	0
1	50,000		
1	25,000		

70,000

Which of the following propositions is correct at the market equilibrium?

a) The consumer surplus is 45,000 francs.

b) The consumer surplus is zero.

c) The consumer surplus will not change if the price increases to 75,001 francs.

d) The consumer surplus is 70,000 francs.

Exercise 9

$X: |\varepsilon^D_{p,X}| = 0.5 \rightarrow$ inelastic

$Y: |\varepsilon^D_{p,Y}| = 0.55 \rightarrow$ inelastic

The price elasticity of demand for good X is $\varepsilon^D{}_{\text{p},X}$ = -0.5, the price elasticity of demand for good Y is $\varepsilon^D{}_{\text{p},Y}$ = -0.55, and the cross-price elasticity of good X with respect to the price of good Y is negative ($\varepsilon^X{}_{p,Y} < 0$).

Which of the following options is correct?

a) Goods X and Y are substitutes and the demand for good X is elastic.

b) Goods X and Y are substitutes and the demand for good Y is inelastic.

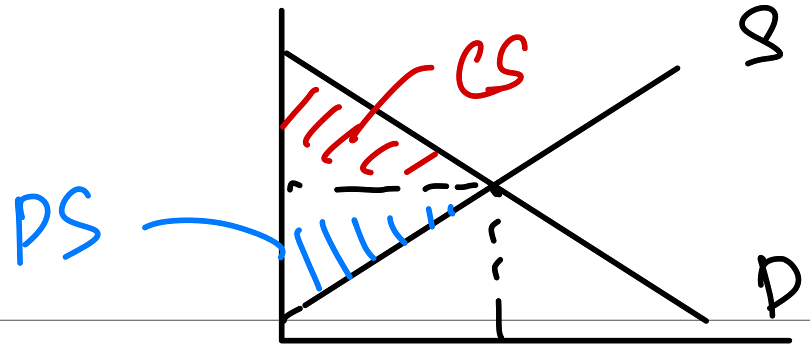
c) Goods X and Y are complements and the demand for good X is elastic.

d) Goods X and Y are complements and the demand for good Y is inelastic.

$\varepsilon^X_{P,Y} < 0$

$\Rightarrow$ X & Y are complements

Exercise 10



Among the following statements, which one is correct?

a) The consumer surplus can be graphically shown by the area below the demand curve and above the firm's marginal cost.

b) Consumer surplus equals producer surplus at equilibrium.

c) The total surplus can be graphically shown by the area below the demand curve and above the firm's marginal cost.

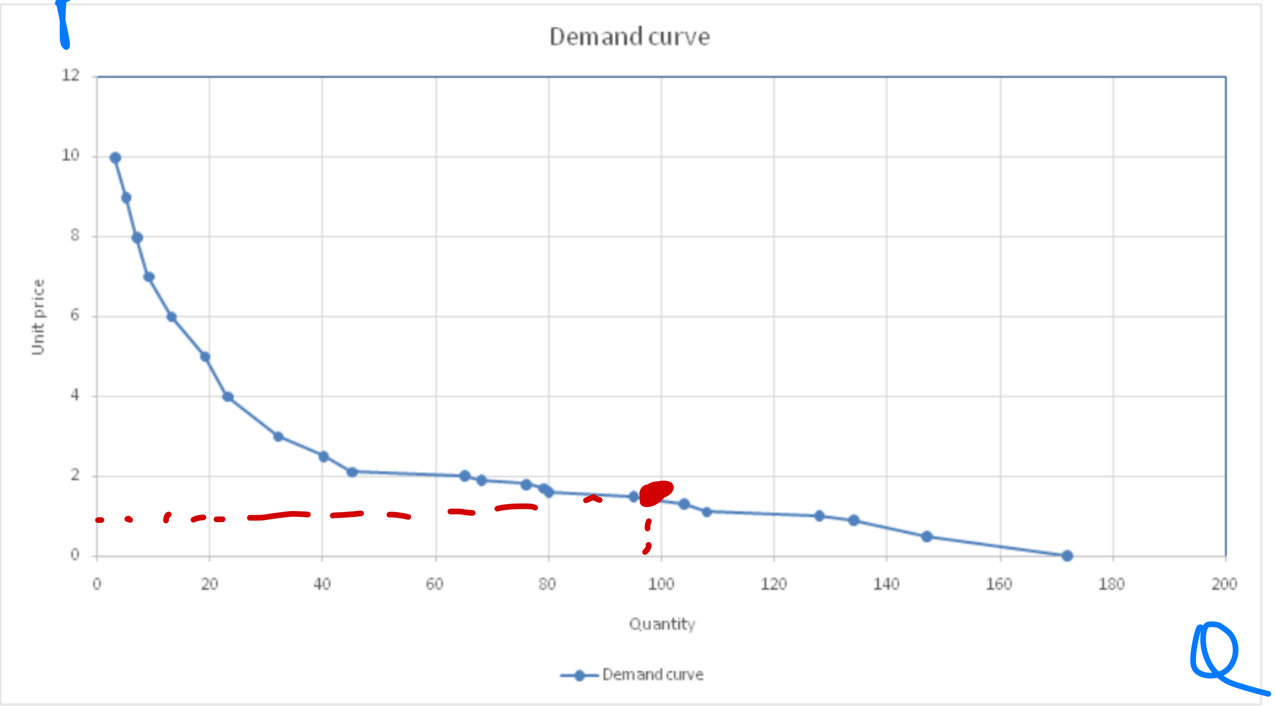
d) The producer surplus is equal to its p

Total revenue = $Q \cdot P$ = TR

Exercise 11

The figure below reports the demand curve of some good. At what combination of price and quantity does the seller maximize their revenue?

P



Q

TR=0 a) At Price = 0 CHF (Quantity demanded is 172).

TR= 73.5 b) At Price = 0.5 CHF (Quantity demanded is 147).

TR=128 c) At Price = 1.0 CHF (Quantity demanded is 128).

TR= 142.5 d) At Price = 1.5 CHF (Quantity demanded is 95).

TR =130 e) At Price = 2.0 CHF (Quantity demanded is 65).